BMC Genomics

RESEARCH

Open Access

Diet of the blue crab (*Callinectes sapidus)* during range expansion in Great Bay Estuary, New Hampshire

Kelsey A. Meyer-Rust[1], Alyssa Strickland[1], Bo-Young Lee[1], Joseph L. Sevigny[2], Gabriela Bradt[3] and Bonnie L. Brown[1*]

Abstract

Great Bay Estuary (GBE), within the rapidly warming Gulf of Maine, has experienced significant ecological shifts this century due to naturalization of invasive species. The range expansion of the American blue crab (*Callinectes sapidus*) currently underway from the mid-Atlantic northward brings the possibility of similar ecological shifts. This study accounts recent trapping and diet analysis of *C. sapidus* in GBE. Diet is an important component of understanding how the blue crab range expansion may affect GBE ecosystem functions. Across all sites and trap types, 27 blue crabs were captured. Metagenomic analysis of shotgun sequencing techniques were used on the gut contents of blue crabs captured. Most specimens had > 50% Eukaryote sequences. Overall results of this gut content study confirm a mixed diet indicative of an opportunistic feeder. Using metagenomics to analyze the diet of blue crabs as they establish viable populations in GBE will be a useful tool for predicting how these range expanding organisms are interacting within this important estuarine ecosystem, which will promote sustainable development by informing end users who may be affected by these crabs to help them meet their needs in the present and future. This project falls within Global Goal SDG14: Life Below Water.

Keywords Range expansion, American blue crab, *Callinectes sapidus*, Metagenomics, Great Bay Estuary, Gulf of Maine

Background

For the past century, the Atlantic blue crab, *Callinectes sapidus* [3], has been expanding its range and invading new areas [4–7]. Concomitant with climate change, species distribution [1, 2] is changing through increases in water temperature and via unintentional introductions. Although species distribution through both climate driven site suitability and human mediated transport interact, they also can be separate paths for expansion. For example, with water temperatures increasing, blue crabs may move as new areas become suitable habitats. This type of climate driven range expansion of blue crabs has been on notice in the Gulf of Maine [4]. The native range of blue crabs extends along the Atlantic Coast of the Americas including Gulf of Mexico [9], and southward to South America [8, 9, 63], with expatriates noted in Maine [10], and northward to Nova Scotia [4, 7, 11]. Blue crab range expansion in the Gulf of Maine has been studied [4] and recently, mated pairs of blue crabs were observed in Great Bay Estuary, New Hampshire [11].

Great Bay Estuary (GBE) is part of the Gulf of Maine region, which is experiencing some the fastest rates of climate change [12–14], resulting in increased sea surface

*Correspondence:
Bonnie L. Brown
bonnie.brown@unh.edu
[1]Department of Biological Sciences, Ecological Genetics Laboratory, University of New Hampshire, 38 Academic Way, Durham, NH 03824, USA
[2]Department of Molecular, Cellular, and Biomedical Sciences, University of New Hampshire, Hubbard Center for Genomic Studies, 35 Colovos Road, Durham, NH 03824, USA
[3]New Hampshire Sea Grant, 15 Strafford Ave, Durham, NH 03824, USA

© The Author(s) 2024. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

temperature [13]. Warmer waters due to climate change can alter marine organism distributions [15], feeding activity [16], interspecific interactions [17], effectiveness of species invasions [18], and range expansion [19]. As this range expanding blue crab, a possible predator of benthic species and prey of pelagic fishes, finds a more hospitable habitat, it is possible that a population of sustained numbers will ensue in GBE.

The arrival of non-native species to New England, specifically GBE, is not a new phenomenon. For example, invasive species such as *Botrylloides sp.* (tunicate) [20], *Carcinus maenas* (European green crab) [20–23], and *Hemigrapsus sanguineus* (Asian shore crab) [20], now are naturalized in GBE, with documented effects [21, 24] and predicted effects [20]. Assuming the blue crab population continues expanding in this unique estuary, there is much interest in its potential impact upon existing ecologically important keystone species in the bay such as oysters [25, 26, & 27] and eelgrass [28, 29, 30, & 31], which could be negatively impacted by blue crabs if they substantially alter the GBE food web. This study provides the first published data for the numbers of blue crabs in GBE, obtained through both bycatch and intentional capture in 2022 and 2023. A snapshot of the diet of those blue crabs was performed using metagenomic analysis of their gut contents [32].

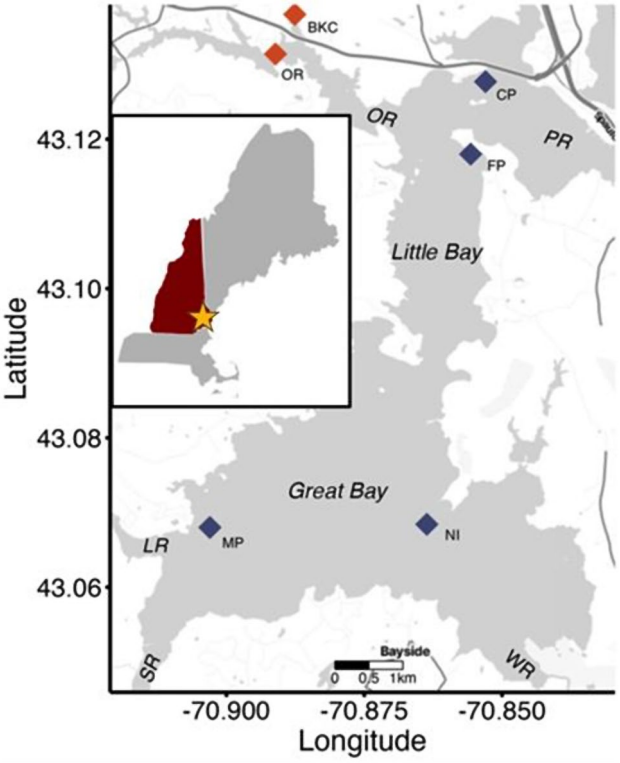


Fig. 1 Map of GBE showing four sites (blue diamonds) where trapezoidal crab traps and crab pots were deployed: CP, FP, NI, and MP, and areas where two crab pots (orange diamonds) were deployed: BKC and OR. For reference, SR: Squamscott River, LR: Lamprey River, OR: Oyster River, PR: Piscataqua River, and WR: Winnicut River are five of the seven rivers that flow into Great Bay

Materials and methods

Blue crab sampling

GBE is a well-mixed estuary located on the southern New Hampshire-Maine border [33]. Beginning in 2020, a trapping system targeting the European green crab (*Carcinus maenas*) was instituted, but only in 2022 and 2023 were blue crabs (*Callinectes sapidus*) captured; therefore, summary of methods and data only for those two years are presented here. In 2022, four sites throughout the estuary were selected for sampling using trapezoidal green crab traps ($n=3$ per site) from Brooks Trap Mill, Maine. The traps measured 0.91 m L × 0.46 m W × 0.28 m H and were made of 4×4 cm coated wire mesh. Nannie Island (NI) and Moody Point (right outside Lamprey River area) (MP) were located near known historical native oyster reefs, and Cedar Point (CP) and Fox Point (FP) were located near known oyster farms. These four sites cover the upper and lower bay areas of GBE (Fig. 1) [34].

During April-November 2022, green crab traps were baited once per week with herring or mackerel placed inside stainless-steel tea bags that lured crabs into the traps but prevented them from consuming the bait, so that they would not skew diet results, and checked twice per week. Although traps were baited (but bait was not available for consumption) and may attract other species along with blue crabs, these data provide an indication of what they would consume in this environment. In collaboration with New Hampshire Sea Grant, starting in July, blue crab pots (0.60 m L × 0.60 m W × 0.5 m H, 4 cm hex mesh) were deployed alongside each $n=3$ trapezoidal trap, also with bait sequestered in tea bags, with the intent to capture blue crabs. All traps were checked at high tide, 48 h after baiting to ensure that the trapped crabs could have had full or semi-full gut contents. This was based on the findings of prior studies that found crabs starved for three days had empty guts [35]; however some studies have shown the blue crab gut system is emptied in 18–24 h [56], meaning most gut contents of blue crabs captured likely were intact when the crabs entered the traps or else were from items consumed during their time in the traps. Blue crab bycatch became a frequent occurrence starting in September and continuing through October (Figs. 2 and 3) in these green crab trapezoidal traps. In 2023, the trapping was repeated with the full intention of catching blue crabs, but with one typical blue crab pot alongside one trapezoidal green crab and one rectangular trap (0.61 m L x 0.23 m W x 0.46 m H) at all sites, checked weekly from April-November. As the previous year, blue crabs again were captured in late summer until early autumn.



Fig. 2 Trapezoidal green crab trap with both blue crabs and green crabs. Nannie Island, 21 September 2022



Fig. 3 Male blue crab caught on 19 October 2022 in a trapezoidal green crab trap at Moody Point-Lamprey River area

Blue crabs also were captured in an Oyster River subtidal zone by NH Fish and Game (NHF&G) in the process of their annual fish seine sampling, and by Great Bay National Estuarine Research Reserve (GBNERR) personnel using blue crab pots deployed in Bunker Creek, intertidal and salt marsh areas. All blue crabs captured were sexed, weighed, and measured (Table 1), and when guts were present (i.e., crabs with rigid exoskeletons; soft crabs that recently completed ecdysis had no gut contents) the gut contents were analyzed for diet using whole-genome shotgun sequencing of DNA extracted from the gut contents. Metadata including latitude and longitude of sites, dates of blue crab sampling, mass and carapace width of blue crabs, and habitat characteristics were compiled into a single table (Table 1).

DNA extraction, sequence acquisition, and statistical analyses

Blue crabs were frozen on the day of capture and subsequently thawed. The gut of each blue crab was excised, and gut contents were pressed into a microfuge tube, about 200-250µl. DNA from blue crab gut samples (all hard carapace adults) was extracted using a QIAGEN DNAeasy PowerSoil Kit® according to the manufacturer's instructions and total DNA was evaluated for quantity/quality using Qubit 4 Fluorometer (Thermo Fisher Scientific) using Qubit 1X dsDNA HS Assay kit (Invitrogen). Sequencing libraries were prepared using the Kapa BioSystems HyperPlus Kit (KR1145 -v3.16). Sequencing was completed on a NovaSeq 6000 with an SP flow cell (paired-end 250 bp reads). Data were demultiplexed using bcl2fastq v2.20.0.422 after which FASTQ files were examined for quality using FastQC v0.11.5 and multiqc v1.11 with default settings [36]. Estimation of duplication rates, adapter sequence trimming, and read merging was completed using fastp v0.23.2 with [37] default settings and the flags '--include_unmerged', '--merge', '-l 50' and '-g' to remove poly-G tails produced by the NovaSeq sequencer.

Translated reads were aligned to the NCBI nr database using the DIAMOND v2.1.9 [38] with BLASTX command with default settings and the '-f' option to write a diamond database file (DAA). Taxonomic and functional classifications were completed using the daa-meganizer tool, implemented in the MEGAN (MetaGenome Analyzer) v6.25.9 software package [39], (see python script in Supplemental Material). Taxonomic classification was performed using the NCBI taxonomy and the GTDB taxonomy [40] and the naïve Lowest Common Ancestor (LCA) algorithm [41]. Classification was enabled with a minimum score threshold of 50.0, a maximum expected value of 0.01, and a top percentage of 10.0 considered for reporting resulting reads being assigned to taxonomic classifications based on read count, using a naive LCA

Table 1 Metadata for mature blue crabs and sampling areas where they were captured

Site	Latitude and Longitude	Date Collected	Sex	Wet Mass (g)	Carapace (mm)	Habitat Characteristics in Sampling Area
Nannie Island	N43.069167 W70.863056	2022-09-02	M**	159	134	Natural/Restoration Oyster Reef Area, Eelgrass
Nannie Island	N43.069167 W70.863056	2022-09-02	F*	145	155	Natural/Restoration Oyster Reef Area, Eelgrass
***Bunker Creek	N43.1367 W70.8876	2022-09-06	M	279.02	188	Salt Marsh
***Bunker Creek	N43.1367 W70.8876	2022-09-06	M	201.1	160	Salt Marsh
Cedar Point	N43.129167 W70.851389	2022-09-12	M	349.79	177	Oyster Farm Area
Nannie Island	N43.069167 W70.863056	2022-09-21	M	147.26	137	Natural/Restoration Oyster Reef Area, Eelgrass
Nannie Island	N43.069167 W70.863056	2022-09-21	M	296.33	162	Natural/Restoration Oyster Reef Area, Eelgrass
Nannie Island	N43.069167 W70.863056	2022-09-21	M	244.63	156	Natural/Restoration Oyster Reef Area, Eelgrass
Nannie Island	N43.069167 W70.863056	2022-09-21	M	216.34	148	Natural/Restoration Oyster Reef Area, Eelgrass
Nannie Island	N43.069167 W70.863056	2022-09-21	M	321.25	172	Natural/Restoration Oyster Reef Area, Eelgrass
Nannie Island	N43.069167 W70.863056	2022-09-21	M	265.16	168	Natural/Restoration Oyster Reef Area, Eelgrass
Nannie Island	N43.069167 W70.863056	2022-09-21	M	252.81	144	Natural/Restoration Oyster Reef Area, Eelgrass
Nannie Island	N43.069167 W70.863056	2022-09-21	M	314.12	161	Natural/Restoration Oyster Reef Area, Eelgrass
Nannie Island	N43.069167 W70.863056	2022-09-21	M	230.7	158	Natural/Restoration Oyster Reef Area, Eelgrass
Nannie Island	N43.069167 W70.863056	2022-09-21	M	233.64	153	Natural/Restoration Oyster Reef Area, Eelgrass
Moody Point	N43.06881 W70.88393	2022-10-07	M**	203.04	145	Natural/Restoration Oyster Reef Area, Eelgrass
Fox Point	N43.116667 W70.854444	2022-10-07	M**	193.98	141	Oyster Farm Area
Fox Point	N43.116667 W70.854444	2022-10-07	F	146.41	147	Oyster Farm Area
Moody Point	N43.06881 W70.88393	2022-10-19	M	184.62	138	Natural/Restoration Oyster Reef Area, Eelgrass
Moody Point	N43.06881 W70.88393	2022-10-19	M	308.28	158	Natural/Restoration Oyster Reef Area, Eelgrass
Moody Point	N43.06881 W70.88393	2023-08-29	F*	148.25	134	Natural/Restoration Oyster Reef Area, Eelgrass
Moody Point	N43.06881 W70.88393	2023-08-29	M	230.9	168	Natural/Restoration Oyster Reef Area, Eelgrass
Moody Point	N43.06881 W70.88393	2023-09-07	M	117.64	116	Natural/Restoration Oyster Reef Area, Eelgrass
Nannie Island	N43.069167 W70.863056	2023-10-03	M	266.01	134	Natural/Restoration Oyster Reef Area, Eelgrass
***Bunker Creek	N43.1367 W70.8876	2023-09-23	M	193.96	126	Salt Marsh
***Oyster River	N43.1314 W70.8911	2023-10-12	M	272.36	154	Natural Oyster Reef, Muddy Bottom
***Oyster River	N43.1314 W70.8911	2023-10-12	F	61.65	87	Natural Oyster Reef, Muddy Bottom

* Denotes post-ecdysis female with no gut contents; not included in diet analysis, ** denotes a male caught in a trap with only a female and *** denotes if not captured in a trapezoidal green crab trap

algorithm with 100.0% coverage and a minimum support threshold of 0.01. Relative abundance normalization [42] and construction of a BIOM formatted table was completed using a custom python script. Data were aggregated to the taxonomic rank of Genus and statistical analyses were performed at the Family level. The method we selected for taxonomic classification is robust. The LCA algorithm has been widely adopted and validated in the scientific community, demonstrating its reliability and robustness across various studies and applications. The LCA algorithm assigns taxonomic classifications by tracing the hits of each read through a taxonomic hierarchy. This allows it to place sequences at the most specific taxonomic level supported by the data, providing a balance between specificity and accuracy. By considering all possible taxonomic assignments and choosing the most common ancestor, the LCA algorithm minimizes errors that can arise from ambiguous or conflicting hits. This conservative approach reduces the likelihood of false positives. In addition, the LCA algorithm is resilient to incomplete or imperfect databases. By assigning sequences to the lowest common ancestor, it avoids overspecification, which is particularly useful when reference databases do not contain exact matches or are hindered with mis-identified sequences. Sequence data are deposited in the NCBI SRA under the BioProject accession PRJNA1090924.

Normalized taxonomic arrays were analyzed using Several-samples Repeated Measures Tests (ANOVA) [43], Tukey's studentized range test [44], and Wilcoxon [45] in PAST4 to examine groupings of dietary consortia by site, date, and crab sex. Principal Components Analysis (PCA) [46, 47] employing the Bray-Curtis dissimilarity index [48, 49] was performed to envision any sample grouping based on diet and to determine which Eukaryotic Families contributed most to differences among samples. Assignment of reads analysis is provided as supplementary material.

Results

Numbers and locations of blue crabs captured in GBE

During 2022 and 2023, the only specimens captured in blue crab pots were those in Oyster River. Across all sites and trap types, 27 blue crabs were captured. In 2022, 18 blue crabs were captured in trapezoidal green crab traps beginning on 2 September and concluding on 19

October, 16 males and 2 females. The first observation of blue crabs was a mated pair captured in a green crab trap deployed on 31 August 2022 at Nannie Island and retrieved 48 h later [11]. The female was immediately post-molt, had shed the gut, and was found to have distended seminal receptacles with sperm plugs [11]. Many crabs (12 of 27) were captured on the degraded oyster grounds at Nannie Island, Moody Point yielded three crabs, two were captured at Fox Point, and one at Cedar Point (Table 1). The largest blue crab captured in a trapezoidal trap was a male from Cedar Point with carapace width of 177 mm. The smallest blue crab was a male with a carapace width of 134 mm at Nannie Island found in a trap with the female (155 mm) with which it had mated. Two additional blue crabs were caught in a crab pot by GBNERR in 2022 in Bunker Creek, both males, one being 188 mm, the largest blue crab captured in this study. In 2023, the first two crabs captured were at Moody Point on 29 August 2023; a mated pair, the female still soft and with sperm plugs [11]. Four additional blue crabs were captured in the trapezoidal green crab traps, three at Moody Point and one at Nannie Island. Of those four crabs, three were male and one was female. Outside the targeted sampling effort, one male blue crab was captured by GBNERR in Oyster River using a crab pot on 7 September, and one male and one female blue crab were captured by NHF&G using a seine in Oyster River on 12 October. Those three were included in the 2023 analyses for a total of 7 blue crabs caught throughout GBE in 2023. The largest 2023 blue crab was a male from Moody Point with a carapace width of 168 mm and the smallest was the female caught at Oyster River with a carapace width of 81 mm. No blue crabs were captured at Fox or Cedar Point in 2023, although farmers reported that blue crabs were seen in these areas.

Metagenomic analysis of diet contents

Metagenomic analysis of shotgun sequencing data from gut contents of blue crabs captured in GBE yielded classified read sets ranging from 207,000 to 2, 900,000 across the 25 samples and identified an average of 590 OTUs per sample. Due to the broad range of reads obtained across gut metagenomes, subsequent statistical analyses used normalized taxon proportions. Eukaryota accounted for an average of 83% of gut assignments (including host), Bacteria for 12% (multiple types), and Virus accounted for roughly 5% (multiple types) of reads (very few Archaea were indicated) (Fig. 4). Excluded from statistical computations were reads assigned to the blue crab family Portunidae (57% of the reads) and the decorator crab family Majidae (16% of reads) as examination of *Genus* assignment indicated these reads were misclassified Portunidae. Another 36% of reads were removed from the statistical analysis because classification was not

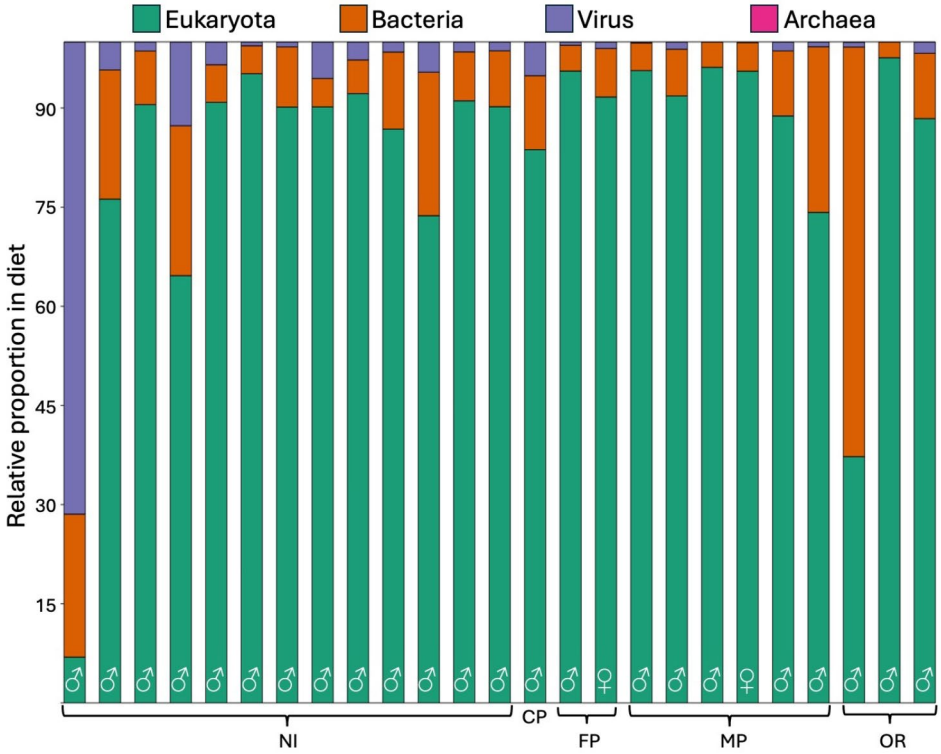


Fig. 4 Stacked bar chart showing relative normalized proportions of eukaryote and other domains detected in guts of adult blue crabs captured in GBE. Most specimens had > 50% Eukaryote sequences. This figure includes all reads

provided to Family. Following these exclusions, between 40,000 and 160,000 normalized reads remained available for each blue crab gut sample. Considering diet components that were observed at ≥0.1% of gut reads (≥92% of guts contained the family), most blue crabs had consumed an array of roughly 24 common organisms (Fig. 5). Bonferroni corrected Wilcoxon test p-values showed no difference between the diets of male and female crabs nor among gut contents across sites or between years. Tukey's pairwise tests indicated that differences among guts were predominantly based on four families that were common to most guts: Enterocytozooa (microsporidian parasite, $p=0$), Penaeidae (shrimp, $p=0$), Nephropidae (lobster, $p=0.0084$), and Pleurotaceae (fungi, $p=0.0085$). Principal Component Analysis illustrated that diets did not cluster by site, year, or sex (Fig. 6). Other families identified at 0.1% or higher were parasites and common estuarine organisms such as mussel, sea slug, amphipod, barnacle, marine worm, acorn worm, horseshoe crab, sea spider, fish, coral [66], and gastropod, as well as several terrestrial species (e.g., crayfish, aphid, spider, pillbug, fly, and butterfly) (Table 2). Organisms identified at ~0.05–0.1% included, sea anemone, oyster, and additional terrestrial species (e.g., snail, beetle, tick) that commonly are found in diets of nearshore benthic organisms (Table 2). Present but at lower proportions were and several teleost families (Supplemental Table 1 Read count summary for all taxa).

Discussion

Although there have been anecdotal reports of blue crabs in New Hampshire waters prior to 2022 [11], this is the first documentation of frequent blue crab capture in GBE. Blue crab range expansion into the Gulf of Maine system due to warmer waters has been predicted [4] and this study adds evidence to substantiate that prediction, as multiple pairs of mating blue crabs were captured over both years as well as consistent catch over both years throughout the Bay (Table 1). Two mating pairs, one in 2022 [11] and another in 2023 (this report), imply that blue crabs inhabiting GBE are reproducing, and as water warms, spawning can occur earlier [50]. Nevertheless, in this study only adults were captured; no juveniles or gravid females were observed. Juveniles and blue crab larvae have been detected in Southern Maine [67], so similar patterns are possible, *albeit* not observed, in this southern region. Validation of a self-sustaining breeding population of blue crabs will require detection of planktonic larvae, as well as observation of gravid females and juveniles.

Evaluating the potential effect of blue crabs on oysters is important because GBE oyster populations have declined precipitously over the past decades [51] due to overharvesting, disease, habitat alteration, and predation. When conceiving of this metagenomic study, species of particular interest were green crabs, oysters, tunicates, and oyster drills, as these organisms are associated with

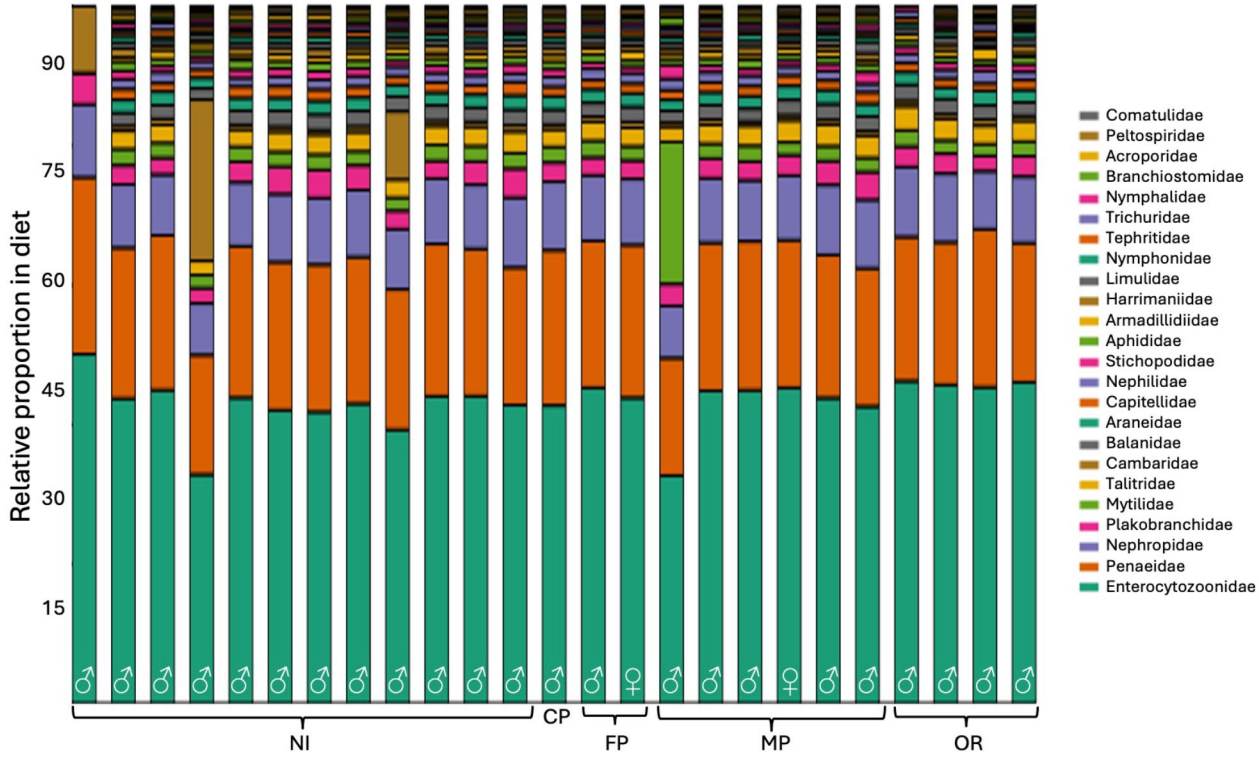


Fig. 5 Relative proportions of the 24 most common (≥0.1% of gut reads) food taxa in diets of GBE blue crabs

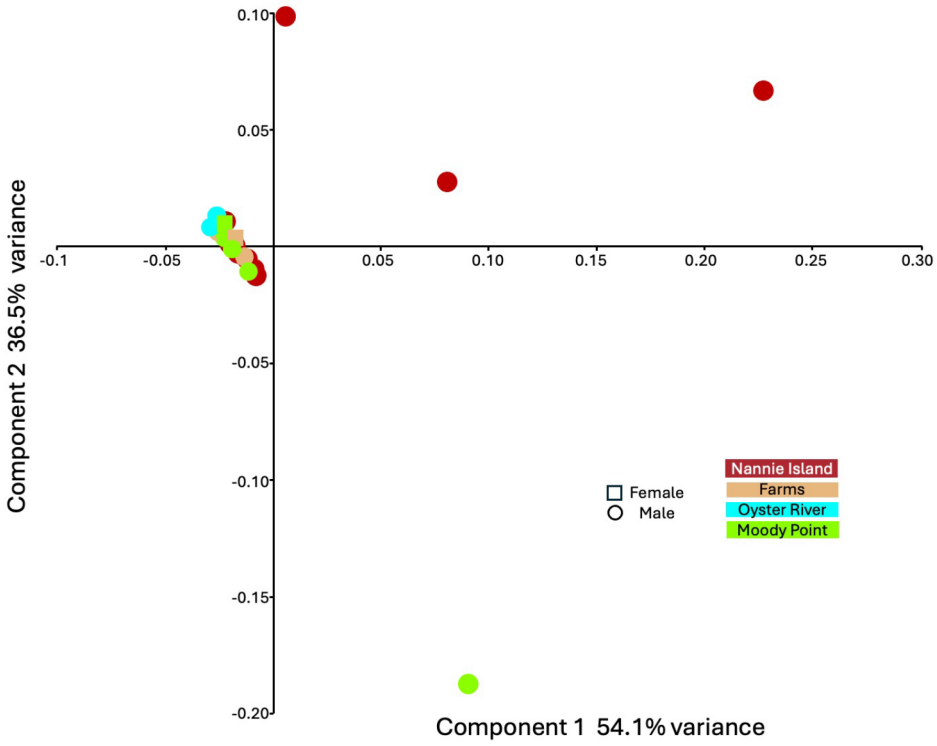


Fig. 6 Principle Components Analysis of blue crab diets based on the 24 most common organisms

oyster restoration in GBE and have been noted as species of concern by the NH Oyster Aquaculture industry [52]. Blue crabs have been known to prey on bivalves [53], and specifically oysters [54, 55] in other studies. Oyster DNA (*Crassoatrea*) was detected in these gut samples, as was the DNA of gastropods (*albeit* not oyster drills) and tunicates.

Green crabs (*Carcinus*) were not detected in any of the blue crab gut samples, even though blue and green crabs were captured in the same traps. Blue crabs have been shown to slow the spread of green crabs and their abundance is less in areas with blue crabs [62]. Thus, although green crabs were not detected in these gut samples, increasing numbers of blue crabs could potentially provide some control of the invasive green crab population, which is thriving in GBE [20–22]. Interestingly, *Callinectes* DNA has appeared in the gut of a green crab captured from the same sampling area as this paper references (data not shown), indicating that some parts of the blue crab, whether they be juvenile, biodeposit, or molt are being consumed by green crabs. The detection of *Callinectes* DNA in the diet of another species provides additional evidence that blue crabs are expanding into GBE and constitutes another approach for tracking the spread of blue crabs. Other arthropods were detected, specifically the family Varunidae (*Hemigrapsus*, another invasive crab species), horseshoe crab (*Limulus*), and lobster (*Homarus*) (Table 2). Lobster is an important fishery in the Gulf of Maine [57], and horseshoe crabs have an abundance of ecological [58] and biomedical importance [59]. Blue crabs could cause ecosystem shifts and might compete with native crabs, such as the rock crab, *Cancer irroratus*, that consume barnacles and mussels [60], species shown here to be consumed by blue crabs (Table 2).

Within Eukaryotes, 9% of sequences were for a microsporidian (*Enterocytozoonidae)*, which is a common parasite of shrimp [64], a common organism in these blue crab gut samples. However, another study [65] described a possible new genus of *Enterospora* within the *Enterocytozoonidae* family that infected two European crab species. Blue crabs may be displaying this parasite as both a mix from consuming infected shrimp or from infection themselves. The proportion of viral sequences (5% of total) reads is not entirely unusual; viruses are ubiquitous and common in aquatic environments [68, 69] and gut samples [70]. Future studies could focus on viruses specifically with this type of methodology because viruses can affect both farmed and wild crustaceans [71], and with blue crabs being a successful fishery elsewhere, it would be important to investigate how these viruses can affect this economically valuable species. Blue crab diets also included insects of terrestrial origin such as spiders and flies. In most cases, the specimens were captured a few meters from shore, so it is highly likely that the diet of adult blue crabs would include terrestrial components associated with runoff.

Table 2 The 45 most common eukaryote families other than crab detected in Great Bay Estuary blue crab (*Callinectes sapidus*) gut metagenomes

Family	Common name	Avg % of gut reads	Number of blue crabs
Enterocytozoonidae	fungal parasite	9.17	25
Penaeidae	shrimp	4.44	25
Nephropidae	lobster	1.97	25
Mytilidae	mussel	0.65	25
Plakobranchidae	sea slug	0.62	24
Talitridae	amphipod	0.58	24
Cambaridae	crayfish	0.51	25
Balanidae	barnacle	0.43	24
Araneidae	spider	0.39	24
Capitellidae	marine worm	0.30	24
Nephilidae	spider	0.29	24
Stichopodidae	sea cucumber	0.23	24
Aphididae	aphid	0.22	24
Armadillidiidae	pillbug	0.19	24
Harrimaniidae	acorn worm	0.18	23
Limulidae	horseshoe crab	0.18	24
Nymphonidae	sea spider	0.16	24
Tephritidae	fruit fly	0.14	24
Trichuridae	parasitic whipworm	0.13	24
Nymphalidae	butterfly	0.12	24
Branchiostomidae	lancelet	0.12	23
Acroporidae	coral	0.12	23
Peltospiridae	gastropod	0.11	23
Comatulidae	crinoid	0.10	23
Aiptasiidae	anemone	0.09	23
Cicadellidae	leafhopper	0.09	23
Ostreidae	oyster	0.09	23
Formicidae	ant	0.09	23
Coccinellidae	beetle	0.08	23
Plutellidae	moth	0.08	23
Hyalellidae	amphipod	0.08	23
Gadidae	cod	0.08	22
Cionidae	sea squirt	0.08	22
Lampyridae	beetle	0.08	23
Dreissenidae	mussel	0.08	23
Ampullariidae	snail	0.07	23
Tenebrionidae	beetle	0.07	23
Trichinellidae	trichina worm	0.07	23
Actiniidae	anemone	0.07	22
Lottiidae	snail	0.06	23
Chrysomelidae	beetle	0.06	23
Ixodidae	tick	0.06	23
Drosophilidae	fly	0.06	23
Liviidae	parasitic hemipteran	0.06	23
Toxopneustidae	urchin	0.05	23

Using metagenomics to analyze the diet of blue crabs as they establish viable populations in GBE will be a useful tool for predicting how these range expanding organisms are interacting within this important estuarine ecosystem. It is important to note, the blue crabs captured were in late August to mid-October, so their diet composition may change depending on the time of year, for example if they were captured in the Spring or mid-Summer. Also, although blue crabs are thought to be opportunistic feeders [61] and their diets may be well-known in other areas, this is the first study to investigate what they are consuming in New Hampshire, which is an area that is new to this species. This study provides a framework for investigating how newly introduced species interact in their new environments, in this case Great Bay Estuary. Although results of this gut content study confirm a mixed diet indicative of an opportunistic feeder, there are organisms found in their guts e.g., lobsters and horseshoe crabs, that are of great economic and ecological importance in this area. Investigating if blue crabs are consuming these types of organisms is important. As more blue crabs enter the Bay due to climate change, this diet composition may change so future studies should continue this work and can use this study as guidance. The potential for blue crab predation is critical information for the growing oyster aquaculture industry in New Hampshire and if the risk is high, farmers will need to adjust their cultivation practices to combat the blue crab. This study provides data that are important for predicting how the range expansion of blue crabs may alter how GBE functions as an ecosystem. This type of research promotes the Sustainable Development Goal 14 of Life Below Water by helping end users, such as shellfish industries, to conserve those important and sustainable species for the present and future.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12864-024-10907-w.

Supplementary Material 1

Supplementary Material 2

Acknowledgements

We thank the UNH Hubbard Center for Genome Studies for their help with data processing, David Shay and others at the UNH Jackson Estuarine Laboratory for their help with field work including boat vessel use and sample/equipment storage, and Caylin Grove and Madison Wilson, part of the UNH EcoGenetics Lab, for assisting on this project. We thank Jason Goldstein (Wells National Estuarine Research Reserve), Chris Peter (Great Bay National Estuarine Research Reserve), Alyson Eberhardt (NH Sea Grant) and the Gulf of Maine Blue Crab Network for their guidance and collaboration of capture methods. We thank Chris Peter, Katharine McGovern, and Heather Ballestro (Great Bay National Estuarine Research Reserve) for blue crab samples from Bunker Creek. We thank Conor O'Donell (New Hampshire Fish and Game) for permits to allow us to conduct this research and for the blue crab samples captured at Oyster River. We thank the NH Oyster Growers for allowing us to trap near their farm areas.

Author contributions

KM conducted field work, data collection, lab work for processing samples collected, and wrote and edited manuscript. AS assisted with field work, data collection, and data analysis and edited manuscript. BYL assisted with lab

work, data analysis, and edited manuscript. JS processed and analyzed data, statistical analysis and edited manuscript. GB provided guidance for field work and edited manuscript. BB oversaw the project, analyzed data, wrote and edited manuscript.

Funding
This study received funding from UNH School of Marine Science and Ocean Engineering, New Hampshire Sea Grant, UNH College of Life Sciences and Agriculture. Partial funding for this study was through the New Hampshire Agricultural Experiment Station [scientific contribution number 3035], supported by the USDA National Institute of Food and Agriculture Hatch Project Numbers 1023564. Bioinformatic analyses were supported by New Hampshire- INBRE through an Institutional Development Award (IDeA), P20GM103506, from the National Institute of General Medical Sciences of the NIH.

Data availability
The data used and analyzed during this study are available from the corresponding author on reasonable request. This sequencing data are deposited in the NCBI SRA under the BioProject accession PRJNA1090924 (https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1090924).

Declarations

Ethics approval and consent to participate
Not applicable, no human subjects were used in this study.

Consent for publication
Not applicable, no human subjects were used in this study.

Competing interests
The authors declare no competing interests.

Received: 29 March 2024 / Accepted: 15 October 2024
Published online: 23 December 2024

References

1. Parmesan C, Yohe G. A globally coherent fingerprint of climate change impacts across natural systems. Nature. 2003;421(6918):37–42.
2. Perry AL, Low PJ, Ellis JR, Reynolds JD. Climate change and distribution shifts in marine fishes. Science. 2005;308(5730):1912–5.
3. Rathbun MJ. (1896). The genus Callinectes. Proceedings of the United States National Museum 18: 349–375.
4. Johnson DS. The savory swimmer swims north: a northern range extension of the blue crab Callinectes sapidus? J Crustac Biol. 2015;35(1):105–10.
5. Daban IB, Cengiz O, Tuncer S. Further range expansion of the blue crab *Callinectes sapidus* (Rathbun, 1896) (Crustacea: Decapoda: Brachyura) in Turkish waters, Northern Aegean Sea: insight into distribution depth. Cahiers De Biologie Marine. 2016;57:175–8.
6. Clavero M, Franch N, Bernardo-Madrid R, López V, Abelló P, Queral JM, Mancinelli G. Severe, rapid and widespread impacts of an Atlantic blue crab invasion. Mar Pollut Bull. 2022;176:113479.
7. Piers H. The blue crab (Callinectes sapidus Rathbun): extension of its range northward to near Halifax, Nova Scotia. Proc Nova Scotian Inst Sci. 1920;15:1918–22.
8. Churchill EP Jr. Life history of the Blue crab. Bull US Bureau Fisheries. 1919;36:95–128.
9. Hay WP. The life history of the blue crab (Callinectes sapidus) (No. 580). US Government Printing Office; 1905.
10. Scattergood LW. Blue crabs (Callinectes sapidus) in Maine. Maine Field Naturalist. 1960;16:59–63.
11. Stasse A, Meyer K, Williams E, Bradt G, Brown BL. First documentation of mating Blue crabs, Callinectes sapidus, in Great Bay Estuary, New Hampshire. Northeastern Naturalist. 2023;30(1):N8–12.
12. Brickman D, Alexander MA, Pershing A, Scott JD, Wang Z. Projections of physical conditions in the Gulf of Maine in 2050. Elementa Sci Anthropocene. 2021;9(1):00055.
13. Pershing AJ, Alexander MA, Hernandez CM, Kerr LA, Le Bris A, Mills KE, Thomas AC. Slow adaptation in the face of rapid warming leads to collapse of the Gulf of Maine Cod fishery. Science. 2015;350(6262):809–12.
14. Pershing AJ, Alexander MA, Brady DC, Brickman D, Curchitser EN, Diamond AW, Wang Y. Climate impacts on the Gulf of Maine ecosystem: a review of observed and expected changes in 2050 from rising temperatures. Elementa Sci Anthropocene. 2021;9(1):00076.
15. Hare JA, Morrison WE, Nelson MW, Stachura MM, Teeters EJ, Griffis RB, Griswold CA. (2016). A vulnerability assessment of fish and invertebrates to climate change on the Northeast US Continental Shelf. PLoS ONE, 11(2), e0146756.
16. O'Connor C, Booth DJ. Response of estuarine fishes to elevated temperatures within temperate Australia: implications of climate change on fish growth and foraging performance. J Exp Mar Biol Ecol. 2021;544:151626.
17. Singer A, Travis JM, Johst K. Interspecific interactions affect species and community responses to climate shifts. Oikos. 2013;122(3):358–66.
18. Jeschke JM, Strayer DL. Usefulness of bioclimatic models for studying climate change and invasive species. Ann N Y Acad Sci. 2008;1134(1):1–24.
19. Sadowski JS, Gonzalez JA, Lonhart SI, Jeppesen R, Grimes TM, Grosholz ED. (2018). Temperature-induced range expansion of a subtropical crab along the California coast. Mar Ecol, 39(5), e12528.
20. Harris LG, Dijkstra JA. Seasonal appearance and monitoring of Invasive species in the Great Bay Estuarine System (2007). PREP Reports & Publications. 65. https://scholars.unh.edu/prep/65
21. Goldstein JS, Morrissey EM, Moretti ED, Watson III, W. H. A comparison of the distribution and abundance of European green crabs and American lobsters in the Great Bay Estuary, New Hampshire, USA. Fish Res. 2017;189:10–7.
22. Fulton BA, Fairchild EA, Warner R. The green crab Carcinus maenas in two New Hampshire estuaries. Part 1: spatial and temporal distribution, sex ratio, average size, and mass. J Crustac Biol. 2013;33(1):25–35.
23. Meyer K, Brown B, Bradt G, Ward K, Brown L, Matso K, Watson W, Goldstein J, Peter C. (2023). Green Crabs – Special Feature (SOOE Extended). *PREP State of Our Estuaries 2023, Extended Version.*
24. Davis RC, Short FT, Burdick DM. Quantifying the effects of green crab damage to eelgrass transplants. Restor Ecol. 1998;6(3):297–302.
25. Langan R. (1997). Assessment of shellfish populations in the Great Bay Estuary.
26. Konisky R, Grizzle RE, Ward KM. Restoring Native Oysters in Great Bay Estuary, NH (2011) (2011). PREP Reports & Publications. 15. https://scholars.unh.edu/prep/15
27. Konisky R, Grizzle R, Ward K, Eckert R, McKeton K. Expanding oyster reefs and populations in Great Bay Estuary, NH. Annual Program Report. University of New Hampshire Sea Grant; 2014.
28. Pe'eri S, Morrison JR, Short FT, Mathieson AC, Brook A, Trowbridge P. Macroalgae and eelgrass mapping in Great Bay Estuary using AISA hyperspectral imagery (2008). PREP reports & publications. 111. https://scholars.unh.edu/prep/111
29. Short FT. Eelgrass Distribution in the Great Bay Estuary 2004 (2006). PREP Reports &Publications. 148. https://scholars.unh.edu/prep/148
30. Short FT. Eelgrass Distribution in the Great Bay Estuary for 2012 (2013). PREP Reports & Publications. 2. https://scholars.unh.edu/prep/2
31. Short FT. Eelgrass Distribution in the Great Bay Estuary for 2013 (2014). PREP Reports & Publications. 348. https://scholars.unh.edu/prep/348
32. Hui TKL, Lo ICN, Wong KKW, Tsang CTT, Tsang LM. Metagenomic analysis of gut microbiome illuminates the mechanisms and evolution of lignocellulose degradation in mangrove herbivorous crabs. BMC Microbiol. 2024;24(1):1–18.
33. Mills K. 2009. Ecological trends in the Great Bay Estuary. Great Bay National Estuarine Research Reserve, Greenland, NH. Document AR-186. 46 pp. Available online at https://www3.epa.gov/region1/npdes/schillerstation/pdfs/AR-186.pdf. Accessed 25 March 2024.
34. Kahle D, Wickham H. Ggmap: spatial visualization with ggplot2. R J. 2013;5(1):144–61.
35. Penney CM, Patton RL, Whiteley NM, Driedzic WR, McGaw IJ. Physiological responses to digestion in low salinity in the crabs Carcinus maenas and Cancer irroratus. Comp Biochem Physiol A: Mol Integr Physiol. 2016;191:127–39.
36. Ewels P, Magnusson M, Lundin S, Käller M. MultiQC: summarize analysis results for multiple tools and samples in a single report. Bioinformatics. 2016;32(19):3047–8.
37. Chen S, Zhou Y, Chen Y, Gu J. Fastp: an ultra-fast all-in-one FASTQ preprocessor. Bioinformatics. 2018;34(17):i884–90.
38. Buchfink B, Xie C, Huson DH. Fast and sensitive protein alignment using DIAMOND. Nat Methods. 2015;12(1):59–60.

39. Huson DH, Weber N. Microbial community analysis using MEGAN. Methods in Enzymology. Volume 531. Academic; 2013. pp. 465–85.
40. Parks DH, Chuvochina M, Chaumeil PA, Rinke C, Mussig AJ, Hugenholtz P. A complete domain-to-species taxonomy for Bacteria and Archaea. Nat Biotechnol. 2020;38(9):1079–86.
41. Huson DH, Auch AF, Qi J, Schuster SC. MEGAN analysis of metagenomic data. Genome Res. 2007;17(3):377–86.
42. Xia Y. Statistical normalization methods in Microbiome data with application to Microbiome cancer research. Gut Microbes. 2023;15:224–4139. https://doi.org/10.1080/19490976.2023.2244139.
43. Girden ER. (1992). ANOVA: Repeated Measures (No. 84). sage.
44. Tukey JW. Comparing individual means in the analysis of variance. Biometrics; 1949. pp. 99–114.
45. Wilcoxon F. Individual comparisons by ranking methods. Breakthroughs in statistics: methodology and distribution. New York, NY: Springer New York; 1992. pp. 196–202.
46. Davis JC, Sampson RJ. Statistics and data analysis in geology. Volume 646. New York: Wiley; 1986.
47. Jolliffe IT. Principal component analysis for special types of data. New York: Springer; 2002. pp. 338–72.
48. Bray JR, Curtis JT. An ordination of the upland forest communities of southern Wisconsin. Ecol Monogr. 1957;27(4):326–49.
49. Doherty KA, Adams RG, Davey N. Unsupervised learning with normalised data and non-euclidean norms. Appl Soft Comput. 2007;7(1):203–10.
50. Schneider AK, Fabrizio MC, Lipcius RN. (2024). Reproductive phenology of the Chesapeake Bay blue crab population in a changing climate. Front Ecol Evol, 11.
51. Grizzle RE, Ward KM. Assessment of recent eastern oyster (Crassostrea virginica) reef restoration projects in the Great Bay Estuary. New Hampshire: Planning for the future. PREP Reports &; 2016. p. 353.
52. Meyer K, Williams L, Ward K, Brown B. (2024) Oyster Grower Perspectives on Green Crab Interactions: Using Applied Social Science to Inform Research and Engagement. *Society and Natural Resources. In Review.*
53. Ebersole EL, Kennedy VS. Prey preferences of blue crabs Callinectes sapidus feeding on three bivalve species. Mar Ecol Progress Ser Oldendorf. 1995;118(1):167–77.
54. Bisker R, Castagna M. Predation on single spat oysters Crassostrea virginica (Gmelin) by blue crabs Callinectes sapidus Rathbun and mud crabs Panopeus herbstii Milne-Edwards. J Shellfish Res. 1987;6(1):37.
55. Eggleston DB. (1990). Behavioural mechanisms underlying variable functional responses of blue crabs, Callinectes sapidus feeding on juvenile oysters, Crassostrea virginica. J Anim Ecol, 615–30.
56. Micheli F. (1995). Behavioural plasticity in prey-size selectivity of the blue crab Callinectes sapidus feeding on bivalve prey. J Anim Ecol, 63–74.
57. Atlantic States Marine Fisheries. 2020.Commission (ASMFC). American Lobster.
58. Botton ML. The ecological importance of horseshoe crabs in estuarine and coastal communities: a review and speculative summary. Biology and Conservation of Horseshoe Crabs; 2009. pp. 45–63.
59. Novitsky TJ. (2009). Biomedical applications of Limulus amebocyte lysate. Biology and Conservation of Horseshoe Crabs, 45–63., 315–329.
60. Sungail J, Brown AC, Alpert K, Maurukas J. Prey selection by Gulf of Maine green crabs (Carcinus maenas), rock crabs (Cancer irroratus) and American lobsters (Homarus americanus): a laboratory study. J Exp Mar Biol Ecol. 2013;449:294–303.
61. Seitz RD, Knick KE, Westphal M. Diet selectivity of juvenile blue crabs (Callinectes sapidus) in Chesapeake Bay. Integr Comp Biol. 2011;51(4):598–607.
62. DeRivera CE, Ruiz GM, Hines AH, Jivoff P. Biotic resistance to invasion: native predator limits abundance and distribution of an introduced crab. Ecology. 2005;86(12):3364–76.
63. Williams AB. The swimming crabs of the genus Callinectes (Decapoda:Portunidae). Fish Bull. 1971;72(3–4):685.
64. Tourtip S, Wongtripop S, Stentiford GD, Bateman KS, Sriurairatana S, Chavadej J, Withyachumnarnkul B. Enterocytozoon hepatopenaei sp. nov.(Microsporida: Enterocytozoonidae), a parasite of the black tiger shrimp Penaeus monodon (Decapoda: Penaeidae): fine structure and phylogenetic relationships. J Invertebr Pathol. 2009;102(1):21–9.
65. Stentiford GD, Bateman KS, Longshaw M, Feist SW. Enterospora cancerin. gen., n. sp., intranuclear within the hepatopancreatocytes of the European edible crab Cancer pagurus. Dis Aquat Organ. 2007;75(1):61–72.
66. Hourigan TF, Cairns SD. (2022) A Comprehensive List of Known Deep-Sea Corals in Waters of the United States and its Island Territories (v.2021).
67. Crane LC, Burke EA, Gutzler BC, Goldstein JS. Evidence of a blue crab (Callinectes sapidus) successfully overwintering in a Southern Maine. Salt Marsh Northeastern Naturalist. 2024;31(3):N11–6.
68. Sime-Ngando T. Environmental bacteriophages: viruses of microbes in aquatic ecosystems. Front Microbiol. 2014;5:355.
69. Wommack KE, Colwell RR. Virioplankton: viruses in aquatic ecosystems. Microbiol Mol Biol Rev. 2000;64(1):69–114.
70. Wang H, Ling Y, Shan T, Yang S, Xu H, Deng X, Zhang W. Gutvirome of mammals and birds reveals high genetic diversity of the family Microviridae. Virus Evol. 2019;5(1):vez013.
71. Dhar AK, Cruz-Flores R, Bateman KS. (2022). Viral diseases of crustaceans. Invertebrate Pathol, 368–99.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.